

Eren Homburg

Software Engineering — Testing — Cloud & Automation

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M.Sc. Computer Science student at ETH Zurich specializing in Secure and Reliable Systems, with hands-on experience in Python, TypeScript, software development, automated testing, cloud infrastructure, and CI/CD workflows. Experienced in developing and operating production software, building automated deployment and validation pipelines, and systematically debugging software and integration issues. Strong interest in reliable software systems, test automation, and applying modern development practices to complex Car-IT environments.

Education

Sep 2025 - Present **M.Sc. Computer Science**
Major: Secure and Reliable Systems
Minor: Data Management,
ETH Zürich, Zürich, Switzerland
- **GPA (current): 5.35**

Sep 2022 - Aug 2025 **B.Sc. Computer Science,**
ETH Zürich, Zürich, Switzerland
- Thesis: 'DBox Plus: An AI-Assisted Code Learning Tutor' supervised by Prof. Dr. April Wang
- **Core Courses: Introduction to Machine Learning, Human-Computer Interaction, Rigorous Software Engineering, Compiler Design, and Visual Computing**

Experience

Nov 2025 – Present **Software Developer,**
ETH Zürich – GenAI-R&D for Future-Proof Teaching, Zürich, Switzerland
- **Skills: Python, React, TypeScript, Backend Integration, Production Software, Testing, Debugging, Documentation**
- Develop and improve **EthelExam**, an open-source **production platform** supporting exam processing, grading, review, and feedback workflows for ETH courses.
- Integrate **React/TypeScript frontends with Python-based backend services** and contribute to **reliable end-to-end application workflows**.
- Improve software reliability through **testing, debugging, documentation, and iterative development**, working across frontend and backend components.
- Collaborate with researchers and instructors to translate requirements into **maintainable technical solutions** used in real course operations.

Jun 2025 – Present **Full Stack Developer,**
ASTAZ – Student Association, Zürich, Switzerland
- **Skills: PHP, MySQL, Deployment, Production Debugging, Third-Party Integrations, Automation, Git**
- Maintain and improve a **production platform serving 4,000+ registered users**, including application, database, payment, and administrative workflows.
- Deploy and operate the platform, **troubleshooting issues** across application logic, hosting configuration, integrations, payments, and data workflows.
- Build **internal tools and automated workflows** while maintaining third-party integrations and the underlying MySQL-based application.
- Take ownership of **production issues** and implement **reliable technical improvements** based on operational needs.

Projects

Feb 2025 – Jul 2025

DBox Plus – AI-Assisted Code Learning Tutor,

Bachelor Thesis supervised by Prof. April Wang, ETH Zürich, Switzerland,

dbox.com 

- **Skills:** **TypeScript**, React, **Python**, LLM APIs, Prompt Engineering, **Cloudflare Workers**, **Docker**
- Built a **full-stack production-oriented web application** with React/TypeScript and backend service integrations.
- Integrated LLM APIs into interactive workflows for AI-assisted guidance, state management, and feedback.
- **Deployed backend services** on Cloudflare Workers with R2, KV, caching, rate limiting, and file streaming.

Feb 2026 – Jun 2026

Automated Data Processing Pipeline for Spatial Reasoning Benchmark,

3D Computer Vision project with Microsoft collaborators, ETH Zürich, Switzerland,

beyond3d-bench.github.io 

- **Skills:** **Python**, Data Processing, **Automation**, **Data Validation**, **Testing**, **Debugging**
- Built **automated data-processing pipelines** transforming 3D object annotations, trajectories, visibility states, and temporal events into structured VLM evaluation tasks.
- Developed **automated validation and analysis tooling** to verify pipeline outputs, **identify systematic errors**, and **debug failure cases**.

Nov 2024 – Dec 2024

Urban AI – Scalable Web Application for Urban Insight,

Fundamentals of Web Engineering, ETH Zürich, Switzerland,

github.com/repo 

- **Skills:** React, **TypeScript**, AI Integration, **External APIs**, **Cloudflare Workers**, **Deployment**
- Built a full-stack web application integrating **external APIs** and deep-learning models to generate insights from user-uploaded images.
- **Deployed backend services** on Cloudflare Workers with authentication, caching, token handling, and **third-party integrations**.

Feb 2026 – Jun 2026

Dynamic Scheduling in Google Cloud Platform with Kubernetes,

Cloud Computing Architecture, ETH Zürich, Switzerland

- **Skills:** **Python**, **Kubernetes**, **Google Cloud Platform**, **Docker**, **Automation**, **Performance Analysis**
- Built **automated deployment and experiment pipelines** on GCP/Kubernetes for latency-sensitive and compute-intensive workloads.
- **Automated performance measurement** and optimized resource allocation while maintaining strict **95th-percentile latency requirements**.

Skills

Programming
Languages

Python, **TypeScript**, JavaScript, PHP, **Java**, C

AI & Automation

LLM APIs, Prompt Engineering, **Workflow Automation**, AI-Enabled Applications, **Scripting**, AI Evaluation

Testing & Quality

Playwright, **Automated Testing**, **End-to-End Testing**, **Debugging**, **Test Validation**, Technical Documentation, **Git Workflows**, Production Software

Web & Backend

React, **REST APIs**, Backend Integration, SQL, MySQL, Authentication, API Integration

Cloud &
Infrastructure

Docker, **Kubernetes**, **Google Cloud Platform**, Cloudflare Workers, **Git**, **Deployment**, **Monitoring**

Software Engineering

Testing, **Debugging**, **Technical Documentation**, Stakeholder Requirements, **Production Software**